

LEARNING INFRASTRUCTURE FOR THE NEXT BILLION CHILDREN

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Programmable physical learning environments where children build,
experiment and develop capabilities with shared AI.

NBC · built by Algo Peers · Sam Quansah, founder · Pre-seed · September 2026

PROBLEM

ANSWERS ARE GETTING CHEAPER. CAPABILITY ISN'T.

AI makes explanations and finished work easy to get. The chance to investigate, build, get it wrong and try again still costs time, materials and skilled adults.

No equipment

A child in Cape Coast has few chances to experiment with real things.

AI everywhere

A child with plenty of AI can't explain or repeat the result when the help changes.

Unseen skill

A child shows real understanding through action, and no test records it.

SOLUTION

PROGRAMMABLE PHYSICAL LEARNING ENVIRONMENTS

Programmable objects

The same blocks, tokens and materials take on new roles and rules, so one kit supports many challenges.

Learning Worlds

Challenges with real consequences: predict, build, test, explain, revise.

Shared AI

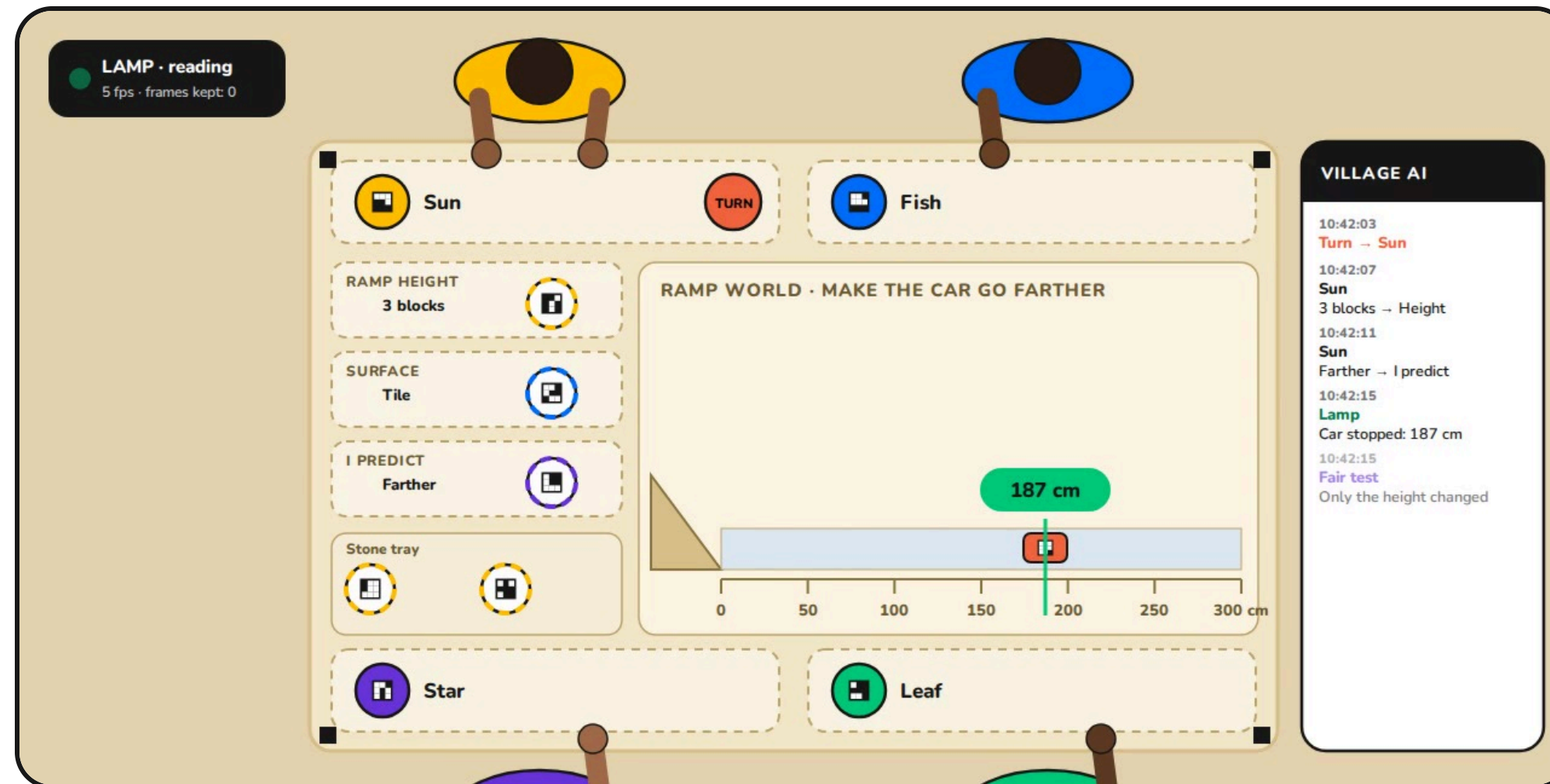
One AI shared by a centre helps facilitators prepare, translate and give feedback. No device per child.

Evidence of capability

What a child can explain, create and do when the task changes. Families can see and correct it.

Learning comes first. The experience has to be worth it for the child. Evidence supports the next learning decision, and a child's record is never the product.

HOW IT WORKS



- 1 The facilitator sets up a world: Ramp World, "make the car go farther".
- 2 Children predict, then change one thing and run it.
- 3 They explain what happened, and try again.
- 4 Later they meet a different world with the same idea.
- 5 Does the skill hold when the task changes? That is the evidence.

Concept image with simulated data. The first version uses printed tokens and a facilitator. Cameras, sensors and AI are added only where they earn their cost.

WHY NOW

THREE THINGS CHANGED

AI access is spreading, unevenly

Some children have more AI than they can use well. Many have almost none. Both need real practice.

Computing became compulsory

Ghana's schools must now teach computing, and most have few working computers.

Shared AI is cheap enough

One phone or computer can now serve a whole learning centre, offline or online.

TRACTION

FIVE YEARS OF PRACTICE. NBC STARTS FROM IT.

Algo Peers, since 2021 · Cape Coast, Ghana

- 1,000+ children reached through our programmes
- Families pay for our after-school programmes
- \$93,511 in grants, including Global Affairs Canada
- 31 public schools assessed (11,214 students)
- Our own learning platform and educator training

NBC, started 2026

- Thesis, three Learning Worlds and a draft task format
- Website and classroom prototypes
- No NBC pilots or revenue yet
- Algo Peers' results don't prove NBC works. The pilot will test that

FIRST CUSTOMER

PROGRAMMES THAT ALREADY RUN HANDS-ON LEARNING

Independent operators of recurring hands-on programmes for children aged about 9–12, starting with reachable operators in Cape Coast.

ROLE	WHO
Uses it	Children in the programme
Runs it	A facilitator
Buys it	The programme owner. Parents may fund it through fees

Offer: NBC Learning Worlds. Reusable physical experiences, facilitator guidance and a short evidence summary that helps choose each child's next step.

ONE CORE BUSINESS, TWO OPTIONS

Core

Learning service

A programme or site fee from operators. A family subscription through our platform is a separate test.

Option

Shared AI service

Useful AI work for operators and communities, only if they pay for it again and again.

Option

Creators and licensing

Licensed experience packages and set-up services, and later, verification.

The learning business must stand alone. Our downside case assumes zero revenue from the options. We never sell children's data.

UNIT ECONOMICS

HOW A SITE BREAKS EVEN

PARTICIPANTS A YEAR	COST PER PARTICIPANT
60	\$26
120	\$16
240	\$11

Example: \$1,200 fixed delivery cost a year plus \$6 per participant. At a price of \$15, a site covers these costs at **134 participants**.

Illustration only, not our price or forecast. Real prices come from the pilot's cost sheet. We publish projections once pilot data supports them.

COMPETITION

WHAT CHILDREN USE TODAY, AND WHAT WE MUST BEAT

	REAL HANDS-ON PRACTICE	WORKS WITHOUT A DEVICE PER CHILD	SHOWS WHAT A CHILD CAN DO
Tablet and AI tutor apps	No	No	Screen only
Robotics and coding kits	Yes	Costly	No
Exams and tests	No	Yes	Recall only
Good materials + a great teacher	Yes	Yes	Informal
NBC	Yes	Yes	Yes

Our real competitor is the last row. NBC has to make great hands-on learning cheaper to repeat and easier to run.

MILESTONES

EIGHT WEEKS TO A CLEAR DECISION

Weeks 1–3

12–16 operator conversations. One capability, three task designs, a costed pilot offer. Offers to 10 qualified buyers.

Weeks 4–6

External facilitators deliver it. A delayed, unfamiliar task, scored blind. Payments and full costs recorded.

Weeks 7–8

A contrasting site if access is ready. Review every gate: repeat, simplify, change payer or pause.

Pass lines, set before we start: 3 of 10 buyers pay · 2 outside facilitators deliver within support · results are readable · scorers agree · each site covers its direct costs.

RISKS

WHAT COULD MAKE US WRONG

- **Substitution:** good materials and a great facilitator may do as well, more cheaply. Then we simplify to the part that helps.
- **Transfer** may not appear, or may cost too much to measure well.
- **Operators** may value the programme but not pay enough to cover delivery.
- **Trust:** automatic skill scoring could lose trust. Records stay human-readable, and NBC works without them.

TEAM

BUILT BY THE PEOPLE ALREADY IN THE ROOM

Sam Quansah

Founder

Founded Algo Peers in 2021. Ed.M. candidate, Harvard GSE. 2024 Mandela Washington Fellow. Former Google Africa Developer Scholarship mentor.

Vera Aninakwah

Learning Designer

Culturally responsive AI, computing and robotics for K-12. BBC micro:bit Champion and Africa Community Lead. Penn GSE scholarship recipient.

Nana Adwoa Nsiah

Learning Experience Designer

Leads our Learning Media Lab: animation and games for African children. micro:bit Champion. NYU Steinhardt scholarship recipient.

Hiring first: a learning scientist, and a technical lead for physical computing and offline software.

HOW CHILDREN WHO CAN'T PAY GET ACCESS

Sponsored schools

Companies and foundations fund Learning Worlds in public schools. Our non-profit delivers them.

Education authorities

License Learning Worlds and teacher training as computing becomes compulsory. After the pilot.

Community centres

Libraries and makerspaces host kits and a shared AI. Local government gives space; sponsors fund kit.

Universities

Co-design and independently evaluate our learning studies.

Two entities: Algo Peers Ltd builds and sells NBC. Algo Peers Impacts, our non-profit, delivers access. Sponsor money is ring-fenced, and children's data is never part of any deal.

THE ASK

RAISING A PRE-SEED ROUND

The pilot

The eight-week programme with external facilitators, delayed tests and paid buyers.

A second site

A contrasting setting, with different access to AI, for the same age group.

The team

A learning scientist and a technical lead, plus a properly powered follow-up study.

Round details on request.